

Qatar-4b

R-2651

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_181124 · created 2026-08-29T23:26:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458446.57727 +/- 0.00069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.053
Transit depth (Rp/Rs)^2	3.37 +/- 1.96 [%]
Orbital Inclination (inc)	87.95 +/- 2.01
Airmass coefficient 1 (a1)	0.69 +/- 0.042
Airmass coefficient 2 (a2)	0.34 +/- 0.36
Scatter in the residuals of the lightcurve fit is	6.13 %
Best Comparison Star	#1 - [524, 43]
Optimal Method	PSF photometry
Transit Duration (day)	0.0597 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.183 ± 0.053 vs 0.138 (deviation 0.61σ)
Duration fit vs published	0.0597 d vs 0.0892 d (deviation 4.94σ)
Mid-time O-C	-1.3 min
Depth SNR	2.5
Residual scatter	6.13 %

Light curve

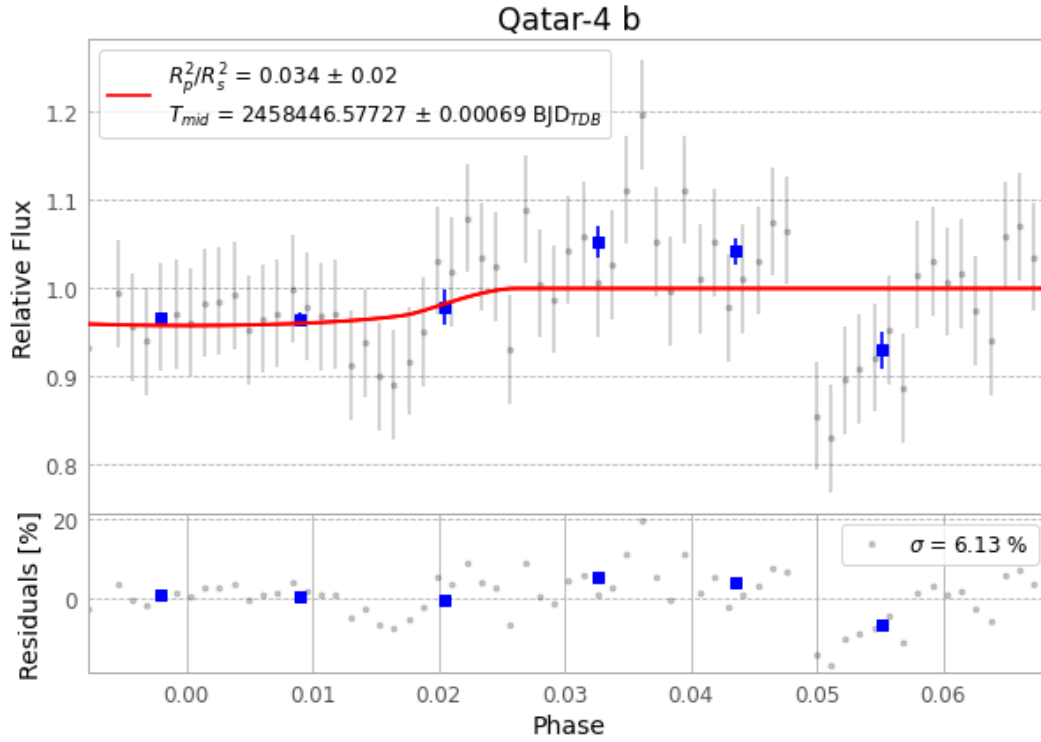


Figure 1 — FinalLightCurve_Qatar-4 b_2018-11-23.png (SHA-256 b0afec46a4085395...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 892cca4d2d827dbb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-4181124012411.FITS → Qatar-4181124044212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-181124.log (7172905c0de5...), FinalLightCurve_Qatar-4 b_2018-11-23.png (b0afec46a408...), FinalLightCurve_Qatar-4 b_2018-11-23.csv (b0ab06180ac7...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:26Z.